



Zero Day Engineering

research & consulting

ADVANCED HYPERVISOR EXPLOIT ENGINEERING

TRAINING OVERVIEW

This 4-day training course is a full-stack exploit development bootcamp targeting hypervisors. It is created and taught by a highly experienced binary hacker who is specialized in hypervisor security research, having established credibility as a successful participant in Pwn2Own competitions in the Virtualization category, reverse engineered and found oday bugs in multiple hypervisor implementations.

Key principle in this course is a systematical, broad and practical coverage of everything that one could possibly need to develop a high-quality exploit for an arbitrary hypervisor, given an exploitable bug. We'll take a look at various attack scenarios: from DoS to Infoleak to RCE and VM Escape, and various classes of security vulnerabilities that are common in hypervisors, with their respective exploitation techniques for both open source and proprietary software. Set of practical exercises will be targeting VMware ESXi.

For best results in the strategic perspective attendees are advised to take this training together with the "[Hypervisor Vulnerability Research](#)" training. Beginners will be assumed to take the latter training first.

AUDIENCE

Primary audience: professional application security researchers and low-level hackers.

System programmers and virtualization engineers will benefit from this course by gaining an advanced attacker's perspective on their code.

Those specialists who are already familiar with hypervisor security will find value in the original methodology and the all-encompassing knowledge exposition of this course.

PREREQUISITES

Mandatory:

- C-code reading & Python basic coding.
- x86_64 assembly and reverse engineering.
- VMware Fusion with nested virtualization.

Recommended (optional):

- "[Hypervisor Vulnerability Research](#)" training.
- Prior experience in exploit development.

OBJECTIVES

Upon completion of this training course students should be able to:

- Write an exploit for any hypervisor, either open source or proprietary.
- Quickly evaluate whether the given bug would be easy to exploit, and the time required.
- Choose the right exploitation technique and exploit engineering primitives for the given bug.
- Avoid reinventing well known exploitation techniques by being familiar with public work.
- Construct original exploits and primitives.
- Understand the relative technical complexities of popular hypervisor implementations with respect to exploit engineering.

LEVELS & CERTIFICATION

Complexity: advanced material.

Certificate grade: ZDE AHE

Next level training: specialized on your target.

TRAINING DETAILS

This 4-day training course is designed to convey a comprehensive system of knowledge about advanced exploit engineering for arbitrary hypervisors and low-level virtualization systems. Attendees will be exposed to the peculiarities of developing exploits for hypervisors, including a broad variety of exploitation techniques for popular hypervisor implementations and bug classes, and the respective practical skills.

Training content is focused on exploit development, given an exploitable bug. That largely excludes vulnerability discovery and threat modeling considerations. The latter vulnerability research topics are covered extensively in our specialized "[Hypervisor Vulnerability Research](#)" course.

Hands-on work is 40-50% of the training, comprised of exercises and labs that represent various stages of engineering an advanced hypervisor exploit. Lectures include both essential theory and a review of specific vulnerability exploitation techniques.

Key principle is to expose systematically a broad spectrum of knowledge points that one is likely to need at some point when writing exploits for hypervisors. We'll take a look at both logic bugs and memory safety issues, various attack scenarios from DoS to Infoleak to RCE and VM Escape, and various classes of vulnerabilities with their respective exploitation techniques. We'll then practice to write an advanced hypervisor exploit targeting open source code (initially) and then proprietary binary code.

Gradual rise of complexity is another principle that is encoded in the course structure. We start from relatively easy bug classes and exploit techniques that don't need advanced shell-coding skills, and wrap up with a theoretical exposition of modern state-of-the-art exploitation trends and emerging topics.

VMware ESXi was chosen to be our practical framework for purposes of exploit development exercises, as the most obscure and technically challenging enterprise-grade hypervisor today. Note, because this course strives to be universal and implementation-neutral, target system internals and reverse engineering will be discussed only to the extent as it is required by the practical case study. Eager advanced researchers are referred to our specialized "Reverse Engineering VMware ESXi" training.

All training materials are original and created by the author specifically for this course. The exploits on which exercises are based were developed by the author herself for prior publications or competitions.



ABOUT THE INSTRUCTOR

Alisa Esage is an independent hacker and founder of Zero Day Engineering. She was the first woman to compete solo at Pwn2Own, where she demonstrated a critical virtual machine escape. She currently works at the cutting edges of the exploit field, between advanced theoretical research and non-trivial exploit engineering for traditionally challenging research targets; such as hypervisors, browsers and basebands.

Alisa is known for her ability to reverse engineer structural patterns behind some of the most mysterious zero-days and architectures, and compress them into practical, high-impact methodologies. Her trainings are used by top-tier researchers, government analysts, and professional red teams worldwide.

PROGRAM AT A GLANCE

DAY 1. WARMING UP WITH LOGIC BUGS

Lectures:

- Recap of theory.

Revise the relevant theoretical concepts and knowledge required to understand hypervisor code. Operating systems, hardware devices, CPU.

- Hypervisor Threat Model.

Generalized hypervisor architecture. Attack surfaces, attack vectors, offensive research trends. Implementation-agnostic models.

- Logic bugs in hypervisors.

Case studies, where to find them, how to exploit.

Practicals:

- Developing an advanced hypervisor exploit, stage 0: vulnerability analysis (heap overflow), evaluation of exploitability, solving reachability.

DAY 3. NON-TRIVIAL BUGS & BINARY CODE

Lectures:

- Working with binary code.

Advanced-level recap of effective reverse-engineering techniques that are helpful when attacking proprietary hypervisors.

- Non-trivial memory safety bugs.

Uninitialized variables, race conditions, TOCTOU.

- Case studies.

Deep technical details of non-trivial security bugs in various hypervisor implementations.

- Exploitation techniques.

Specific technical algorithms and primitives that work for non-trivial bugs in hypervisors.

Practicals:

- Stage 2: adjusting proof-of-concepts to proprietary code of the target.

DAY 2. SIMPLE OVERFLOW BUGS & OPEN SOURCE CODE

Lectures:

- Recap of memory safety concepts.

Classes of bugs, general considerations of exploitability by bug class, hypervisor specifics.

- Analyzing open source code.

Familiarizing with large code bases, reverse-engineering target architecture, finding bugs.

- Case studies.

Deep technical details of memory safety issues in various hypervisor implementations.

- Exploitation techniques.

Specific technical algorithms and primitives that work for memory corruption bugs in hypervisors.

Practicals:

- Developing an advanced hypervisor exploit, stage 1: create simple proof-of-concept codes for the target vulnerability.

DAY 4. STATE OF THE ART

Lectures:

- Future of hypervisors.

Rustlang security essentials. Overview of hypervisors written in Rust. Hardware assist technology updates. Technological trends.

- Speculative execution bugs & Rowhammer.

Vs. the Hypervisor Threat Model. Known bugs that affect hypervisors. State of the art.

- Non-trivial exploits.

Principles of exploiting hard bugs & creating non-templated exploits.

Practicals:

- Stage 3: RCE PoC exploit & shellcode. Evaluating further work.

Note: the program may change.

TRAINING PACKAGES & FEES

This training is available live: online and in-person.

Online training is based on a modern streaming platform with high-quality audio and video equipment, and a group chat. The instructor will be available for questions, feedback and technical support for students with Advanced package. Lab setup is DIY.

Self-paced training offers exactly the same learning experience as online training, less the coaching factor (you'll have to time yourself) and quick availability of the instructor's feedback and answers.

All training packages were specifically optimized for online teaching.

LIVE ONLINE TRAINING

Basic package

What's included:

- Access to public online training.
- Instructor's feedback by email.
- Training slides and supplementary materials.
- Training completion certificate (upon request).

Price: €3,900.- NET per person.

Advanced package

What's included:

- Everything in the Basic package.
- Personal feedback and technical support from the instructor while in the class.
- Possibility to get a distinction certificate by demonstrating abilities and successful apprehension in the class.

Price: €4,200.- NET per person.

Limited number of seats.

SELF-PACED COURSE

Basic package

What's included:

- Video lectures, exercises, and walk-through.
- Training slides and supporting materials.
- Access for 3 months.

Price: €2,500.- NET per person.

Complete package

What's included:

- Everything in the Basic package.
- One month of technical support by email.
- One on-demand consultation call with the founder.
- Training completion certificate.
- Lifetime access.

Price: €3,900.- NET per person.

Note: certificates will be available upon request.

PRIVATE & CUSTOMIZED TRAINING

Private and customized training may be available for groups of at least 10 persons.

PUBLIC TRAINING DATES & BOOKING

Refer to our [schedule](#) for the dates of the nearest public training.

This training is available exclusively for direct customer bookings through our website zerodayengineering.com and email. We may refuse to accommodate unauthorized third party bookings.

Bookings of public training seats are accepted via website checkout system. You may [contact us](#) directly for customized and private training bookings, direct bank payments and alternative payment methods.

INDUSTRY FEEDBACK - ZERO DAY ENGINEERING TRAININGS



Jael Koh · 1st
OSEE | OSCE3 | ZDE VR | Corelan Advanced | WISE |
HackSys AKE
7mo · Edited · 🌟

I'm excited to start off 2024 by completing [Zero Day Engineering](#)'s Zero Day Vulnerability Research course.

[Alisa Esage](#) does an amazing job of taking something as complex and intimidating as Vulnerability Research and breaking it down into a comprehensive yet approachable curriculum.

One of my biggest takeaways from the course was Alisa's methodology for finding zero days in modern software. She teaches a systematic approach, using models of vulnerabilities, exploits, and application threats.

For example, by modeling the subsystems and threats of an application, I learned how to prioritize which subsystems to target and discover new conceptual attack vectors.

This course was both enjoyable and incredibly informative. I felt significantly better prepared for vulnerability research by the end, and I believe this systematic approach will help me avoid common pitfalls.

Overall, the course offers a powerful, systematized methodology that I haven't encountered before. It's also a great foundation for beginners interested in entering the field of Vulnerability Research.

I thoroughly enjoyed this course and am eagerly looking forward to its major upgrade later this year.



Peleg Hadar 🐦
@peleghd

The Hypervisor Exploitation One training by [@alisaesage](#) was simply incredible. Alisa is a great teacher with a lot of experience in the Vuln research field.

The Training contains practical exercises, it provides the knowledge you'll need for starting a Hypervisor vuln research!



Mohammad Hussam Alzeyyat · 2nd
Vulnerability Researcher/Cyber Security (Instructor, Labs Developer, Upwork Top Rated)/OSCE, OSMR
2mo · Edited · 🌟

I just finished the "Masterclass: Hacking open source fuzzers for smarter bug hunting" from [zerodayengineering.com](#) by Alisa Esage.

The 4-hour training is an eye-opening to more advanced fuzzing and it's more oriented to learn how to customize the fuzzer for your own purpose and not to learn how to fuzz.

The masterclass focuses on AFL and WinAFL since those are the most powerful frameworks for fuzzing and explains the anatomy of fuzzing and the differences between the old generation fuzzers methodology and the new ones.

I don't think the masterclass is for beginners as much as for intermediate people or those with fuzzing experience.



Josh Pitts 🐦
@ausernamedjosh

The JavaScript Engines for Hackers by [@zerodaytraining](#) was great. Shortened the time to get familiar with V8. Thx [@alisaesage](#)!



Gal Z 🐦
@0xgalz

What a blast! Just finished [@alisaesage](#) 3 days deep technical Hypervisor Vulnerability Research course. Incredible and well designed with high quality materials.
Thanks for everything! Now, off to look at some VMM code \o/



Carlomagno A. · 2nd
Security Researcher
6mo · 🌟

My review for [#ZeroDayVulnerability](#) course by [Alisa Esage](#).

Alisa's does an amazing job at explaining more than decade of wisdom, insights and knowledge on a complex topic as Vulnerability Research in a 4 days course. The value of this information and training goes beyond just specific attack vectors or simplistic how-to's, it covers deep technical hands on training on real world vulnerable applications.

The segment of the course that I enjoyed the most was the establishment of the correct/appropriate mindset, definition of abstract models/frameworks, cognitive skills needed to better improve, this specific points matched with a well structured self-study roadmap will allow you to track your progress by establishing a feedback loop on the concepts that you need to improve upon to excel at Vulnerability Research.



rui 🐦
@fdiskyou

Alisa's training completely changed my mindset and the way I was looking at hypervisors. I highly recommend the Advanced Hypervisor Exploitation training. Actually, if you're interested in hypervisors it's mandatory.



Kapil Khot · 1st
Security Consultant/Penetration Testing | CoreLAN Exploit Dev | OSCP
7mo · Edited · 🌟

Here's my review for [#ZeroDayVulnerabilityResearch](#) class offered by [Zero Day Engineering](#). It's an intensive four days training where [Alisa Esage](#) shares some important tips from her Pwn2Own competition and decade long zero day research experience.

She walked us through real world application's code base to demonstrate how to identify its subsystems and which one to prioritize for testing. Additionally, she walked us through some of the real world vulnerable code snippets from various enterprise and real-world applications to demonstrate several bug classes such as memory corruption vulnerabilities, hardware bugs, and logic bugs etc. Furthermore, she also touched base on the security weaknesses of one of the memory safe languages and shared research tips on upcoming technologies.
One of the modules, Fuzzing Masterclass took us through the internals of coverage guided fuzzers and offered guidance on customizing them for applications not supported out of the box.

Overall, this course helps you build a zero day research mindset, guides you on what needs to be done, and forces you to figure out the most of the "how-to" part, which in my opinion is the best thing.



Mario Romero Serrano · 2nd
Senior Vulnerability Researcher - Cryptography Research
Centre at Technology Innovation Institute (TII) | Cryptograph...
5mo · 🌟

The last few months I have been enjoying and learning about vulnerability research from one of the best trainings available today.

The 4-day Zero Day Vulnerability Research course from [Zero Day Engineering](#) by [Alisa Esage](#) is not only exceptional on a technical level, but it also teaches you a methodology to follow, and personally, what for me is most valuable and concentrates Alisa's years of experience and knowledge: it introduces you to the mindset necessary for your vulnerability research to be successful.

I am already applying the knowledge I have acquired and I am eager to continue learning from her. I'm sure I'll be taking some more of her courses. For now, I will enjoy the course for the second time.

Don't miss the opportunity to do it:



Jacon Walker 🐦
@call_eax

Big thanks to [@alisaesage](#) and [@zerodaytraining](#) for their top-notch training content! Just wrapped up a refreshing course with promising results - from bug discovery to exploitable vulnerability in just 9 days. Dive into my chronicle here: <https://t.co/PCfqzCIs6B>



Veer Singh · 1st
Security Practitioner
7mo · 🌟

I recently (2023) completed the "Hypervisor Vulnerability Research" course by [Alisa Esage](#), which provided a comprehensive exploration of virtualization technology and its security aspects. The course began with a clear introduction to virtualization technology, distinguishing between various types of hypervisors. We then explored the rich functionality of guest services, understanding their practical implementation and potential security challenges. The module on attacking guest services equipped us with valuable perspective on attack vectors, and the in-depth look at VirtualBox's implementation was insightful. Overall, this course is highly recommended for both beginners and experienced professionals looking to enhance their virtualization security skills. Here is a link to the course for anyone interested - <https://lnkd.in/g5N2gNmT>

Training: Hypervisor Vulnerability Research
[zerodayengineering.com](#)



scryh 🐦
@scryh_

Just finished the [@zerodaytraining](#) hypervisor vulnerability research offline course by [@alisaesage](#). Impressive research bundled into very well structured lessons, which are communicated in a clear and comprehensible way. Learned a lot. Thank you very much!
<https://t.co/iW3UAGBUuq>

APPLICATIONS & INQUIRIES

E-mail: contact@zerodayengineering.com.

Public cohorts and booking: zerodayengineering.com

Socials: [@zerodaytraining](#).